



LAB 2

Data Acquisition techniques

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Strategies for data acquisition

Two general methods, with different characteristics and precision, can be used to determine the position of a point on the physical surface of the earth:

Stand-alone (absolute)

- **Pseudo-range** observations;
- Code based;
- Absolute precision (95%): 5-10m;
- Useful for navigation applications, not for topographic one.

Differential (DGPS/RTK)

- Observations (**pseudorange or carrier phase**) are corrected with **differential corrections** calculated in a "base" station of known position;
- Corrections transmitted (via radio, GSM or Internet) and applied in real-time according to the RTCM protocol.

Observations	Distance between stations	Precision
Code-based	A few hundred km	> meter
Phase smoothing	A few hundred km	< meter
Phase-based	10-20 km	Centimeter



Strategies for data acquisition

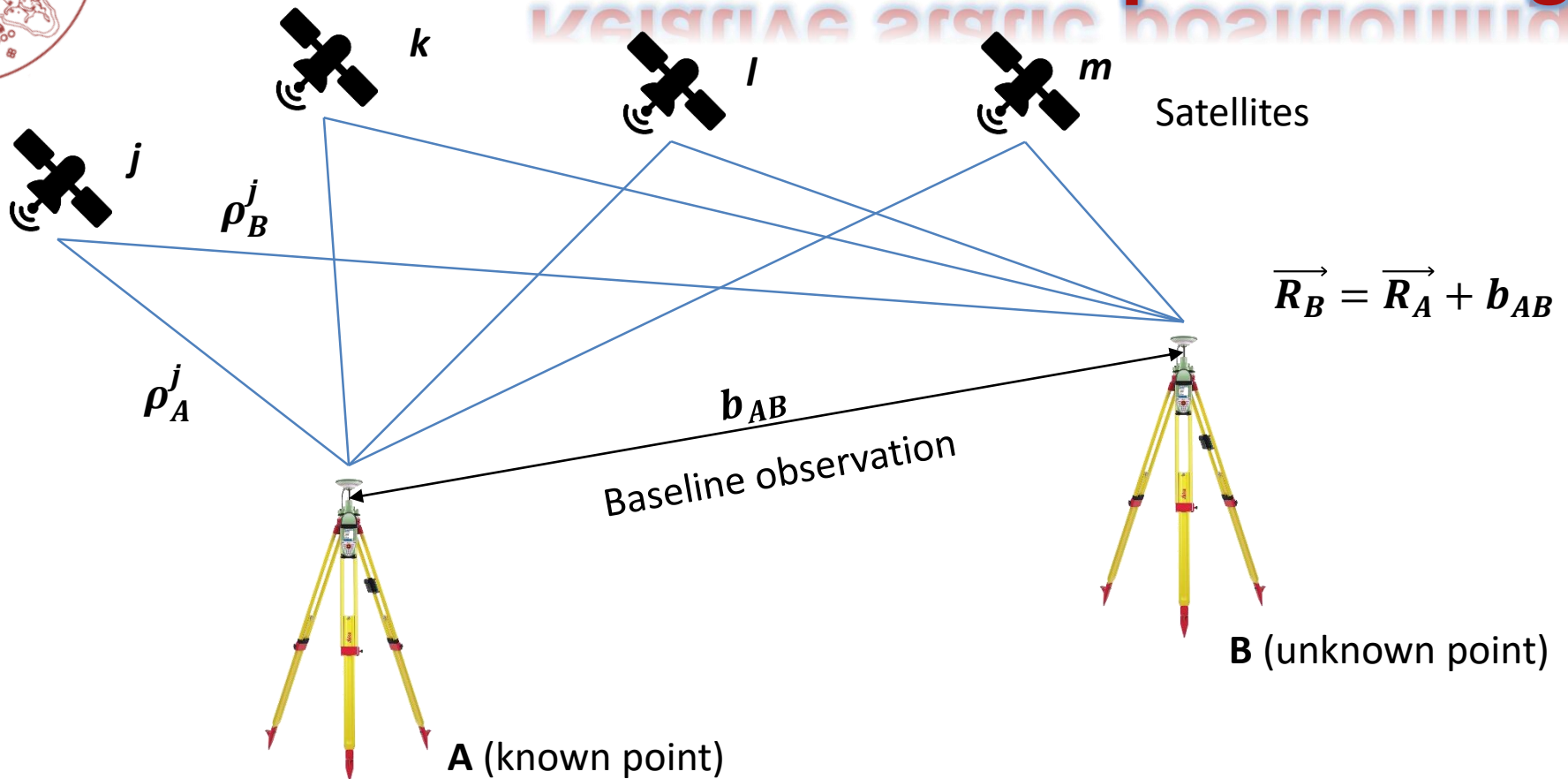
Relative

- the **baseline** vector between the vertices occupied by the stations is determined. The processing is done **a posteriori**, normally on the **phase data**, with the double difference method. The main methods are:

Method	Measur. time	Base-lines	Preci-sion	Rate [sec]	Note
Static	> ½ h 1 h 3÷4 h variable	10 km 20÷30 km >100 km	$10^{-6} \div 10^{-8}$	15÷60	Double frequency if with bases > 20 km
Rapid-static	20÷30 min (L1) 6÷8 min (L1+L2)	<10÷15km	10^{-6}	1÷15	It needs a good satellite configuration
Kinematic Stop and go	< 1 min (at least two epochs)	A few km	Centimeter	1÷5	Requires continuous contact with satellites. Initialization: ➤ Up to 30 min → L1 ➤ 5÷6 min → L1+L2 ➤ On the fly → L1+L2
Kinematic continuous	Continuous	A few km	Centimeter	1÷5 20 Hz	Same as stop and go



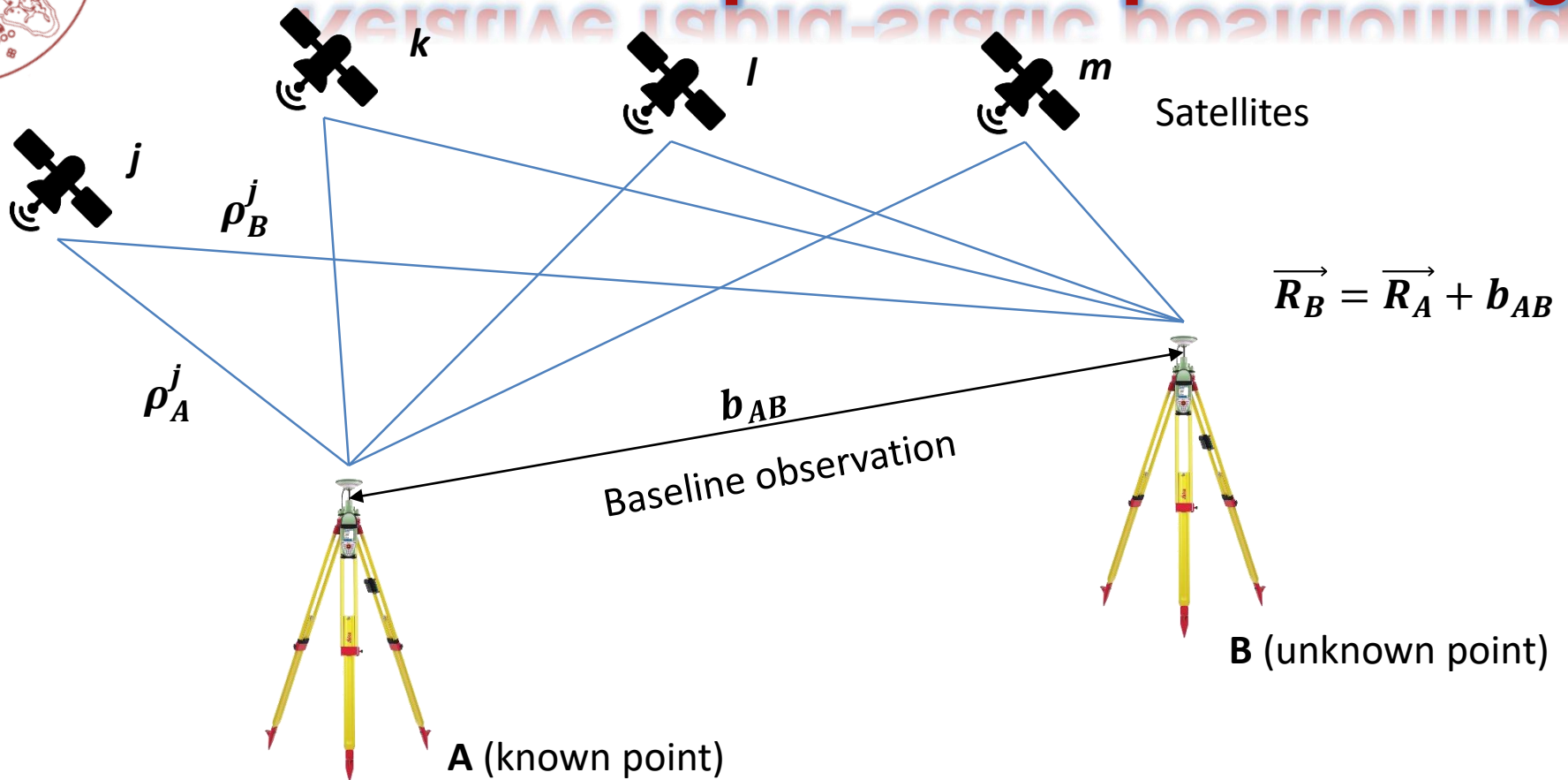
Relative static positioning



Method	Measur. time	Baselines	Precision	Rate [sec]	Note
Static	> ½ h	10 km	$10^{-6} \div 10^{-8}$	15 ÷ 60	Double frequency if with bases > 20 km
	1 h	20 ÷ 30 km			
	3 ÷ 4 h	> 100 km			
	variable				



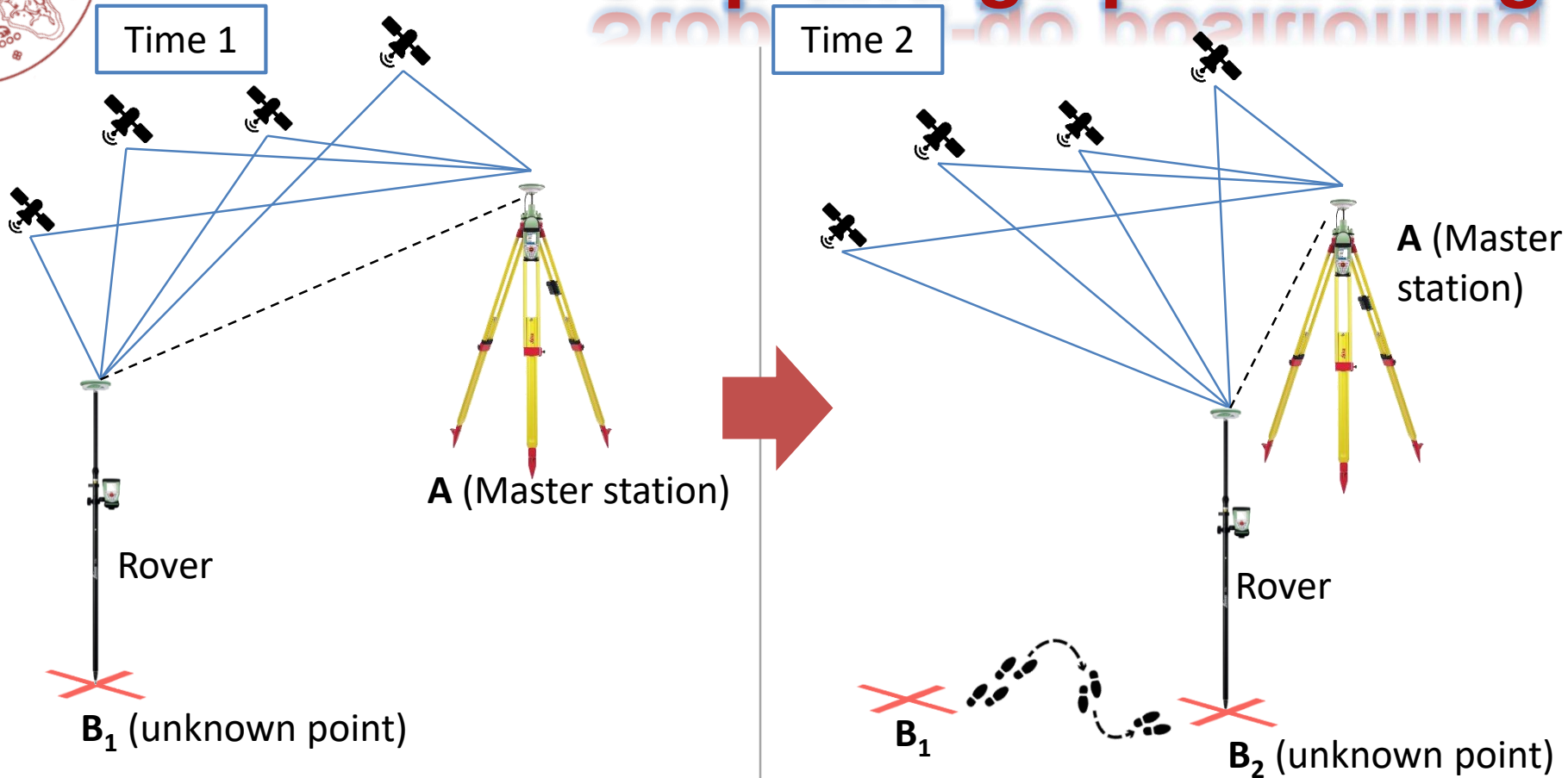
Relative rapid-static positioning



Method	Measur. time	Baselines	Precision	Rate [sec]	Note
rapid-static	20÷30 min (L1) 6÷8 min (L1+L2)	<10÷15km	10^{-6}	1÷15	It needs a good satellite configuration



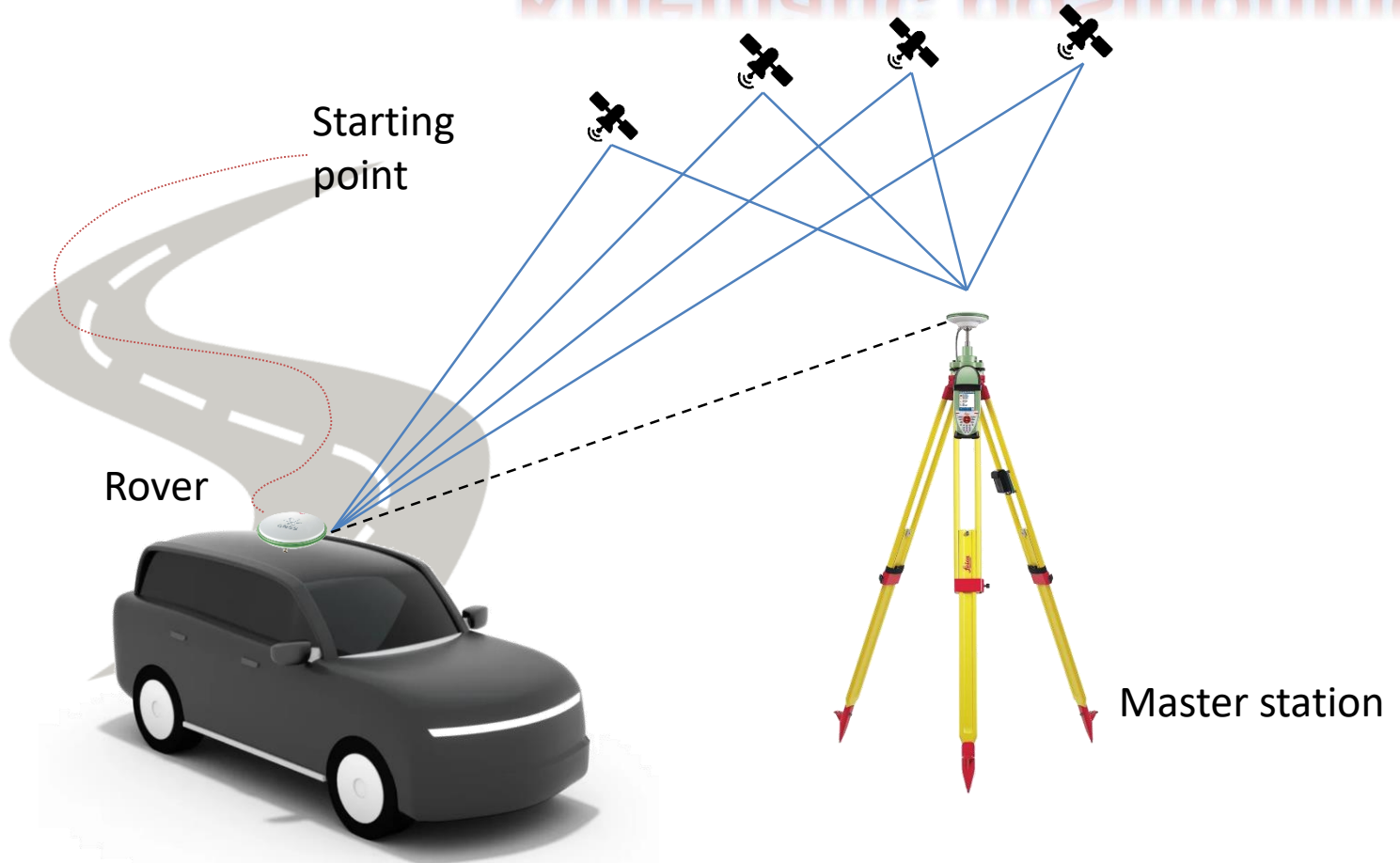
Stop-and-go positioning



Method	Measur. time	Baselines	Precision	Rate [sec]	Note
Kinematic Stop and go	< 1 min (at least two epochs)	A few km	Centimeter	1÷5	Requires continuous contact with satellites. Initialization: ➤ Up to 30 min → L1 ➤ 5÷6 min → L1+L2 ➤ On the fly → L1+L2



kinematic positioning



Method	Measur. time	Base-lines	Precision	Rate [sec]	Note
Kinematic continuous	Continuous	A few km	Centimeter	1÷5 20 Hz	Same as stop and go



Equipment for GNSS positioning

MASS MARKET OR CODE ONLY

- Acquire only C/A component
- > 6 channels (usually > 24)
- limited size
- coordinates are directly shown on the screen

Application: Navigation, approx values, etc



Accuracy	Cost
5÷10 m – Stand-alone Pos. 1÷3 m - DGPS	50-300€



SINGLE FREQUENCY (L1 only)

- Acquire C/A and L1
- > 12 channels
- able to store «RAW data»

Application: real time or static positioning

Accuracy	Cost
mm-cm (short baselines)	500-5000€



Equipment for GNSS positioning



MULTI FREQUENCY P-code

- It can acquire all parts of the signal (L1, L2, C / A, P, L5...).
- In the absence of Anti Spoofing (A-S), the P code can be obtained directly by correlation
- Multi constellations (GPS, GLON, GAL)

Use: all types of positioning (static, kinematic), including precision, without limits on the length of the bases. Particularly suitable for kinematic techniques with On The Fly (OTF) initialization or precision measurement of long bases

MULTI FREQUENCY Y-code

- They have access to the P code even in the presence of A-S

Access to the P code is obtained with an Auxiliary Output Chip - AOC on each channel

- AOC also corrects Selective Availability (SA)

Department of Defense (DoD) authorized access

Double or + frequency multiconstellation receivers

Accuracy	Cost
2÷20 cm in RTK	> 6000€



GNSS data formats

THE GNSS RECEIVER CAN PROVIDE INPUT / OUTPUT:

Pseudorange and carrier phase data (for post-processing)

manufacturer's proprietary format (binary)

Independent **Receiver INdependent Exchange (RINEX)** format (ASCII):

Observations file: **.OBS**

Navigation file with ephemeris: **.NAV**

Real-time coordinates: NMEA (National Marine Electronics Association) protocol via serial port

RTCM (Input/output) for real-time: differential code or phase corrections (Radio Technical Commission for Maritime [Service]).



GNSS integration with other sensors

The GNSS receiver often controls or is controlled by other sensors:

- INS
- Laser scanner
- Photogrammetric cameras
- Atomic clocks
- Geophysical equipment
- Navigational instruments
- Etc...



- Example: provides the trigger pulse of a photogrammetric camera and record the instant of taking.

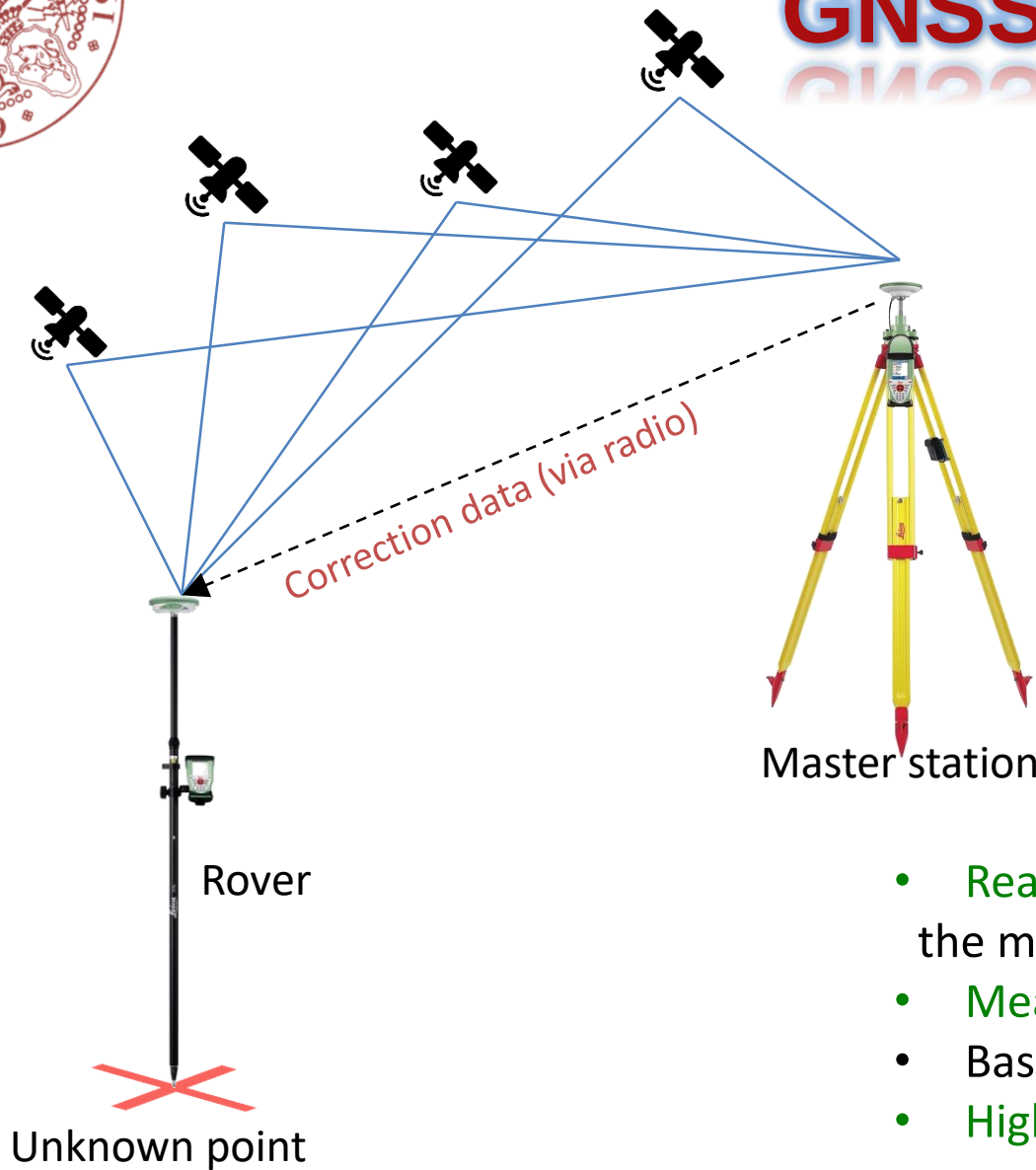


In some receivers are implementable:

- 1PPS: output impulse towards the external instrument
- EVENT MARKER: input impulse (record the instant when the measurement took place in an external instrument)



GNSS RTK positioning



Equipment: at least two **dual frequency geodetic receivers**

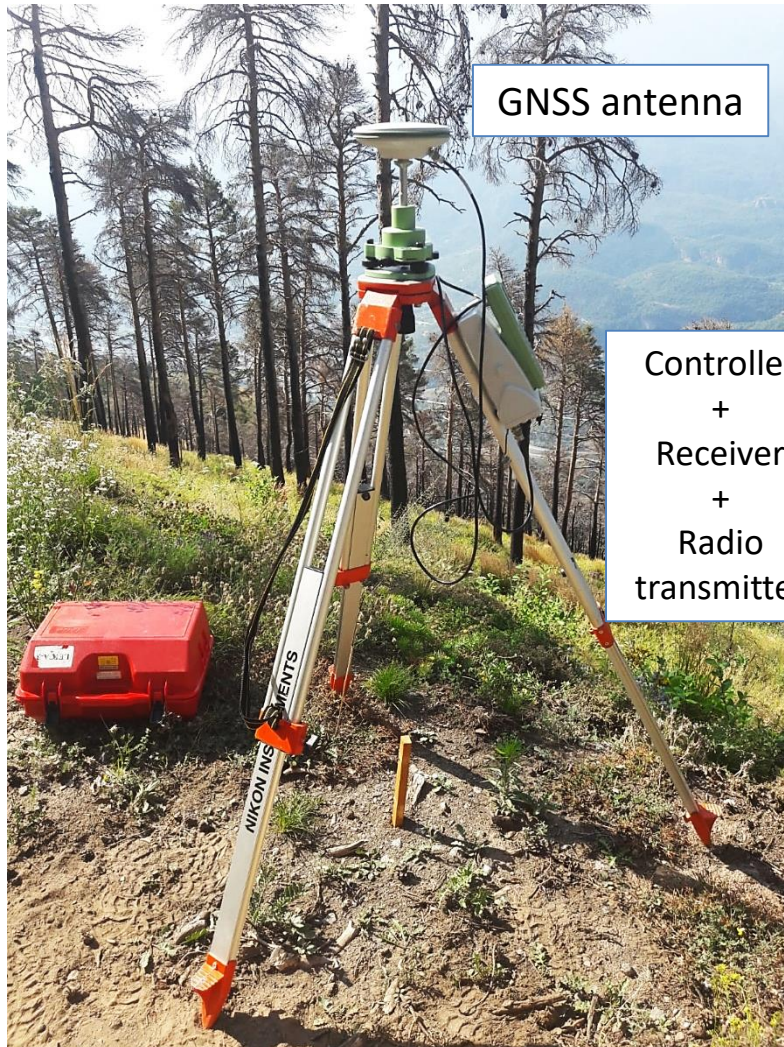
MASTER-ROVER configuration

- **Real time solution**: the coordinates of the master station are known a priori
- **Measurement time**: 5-10 s.
- Baselines: <20km
- **High productivity**



MASTER STATION

GNSS RTK positioning



GNSS antenna

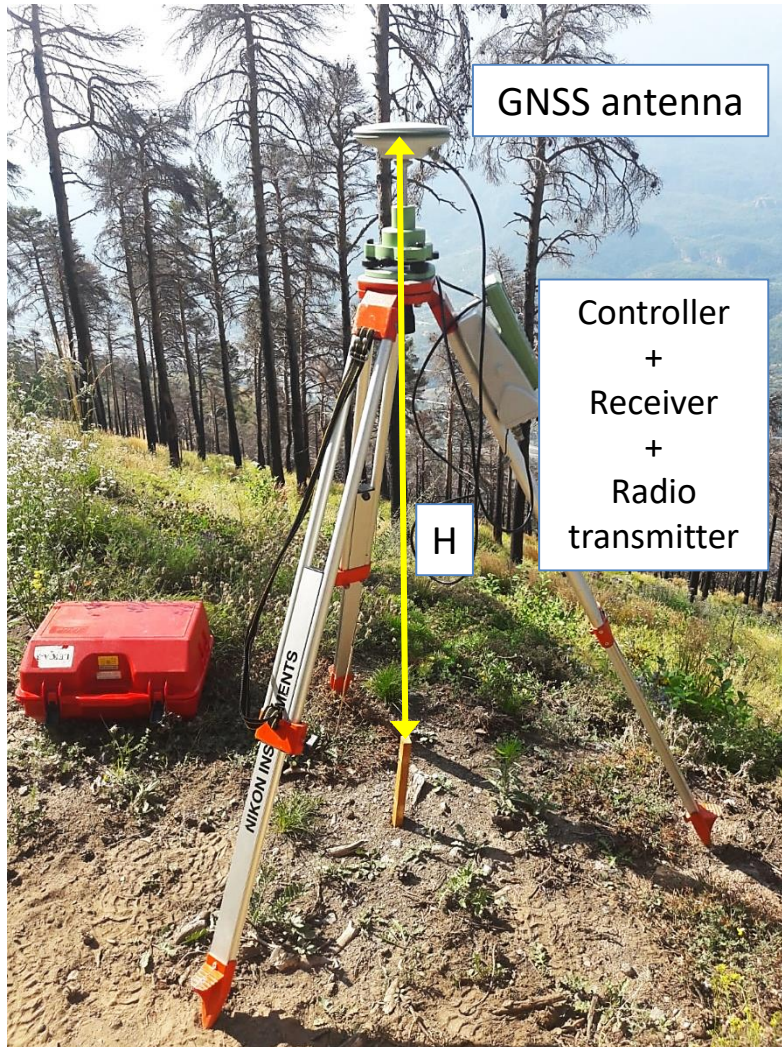
Controller
+
Receiver
+
Radio
transmitter

Operational steps

- Set up the tripod over a point;
- Measure the instrument height;
- Connect the GNSS antenna to the receiver;
- Connect the radio transmitter to the receiver and select the correct frequency;
- Start the receiver and choose the correct job settings;
- Set the known coordinate of the point;
- Start the data acquisition/transmission.



MASTER STATION



GNSS RTK positioning

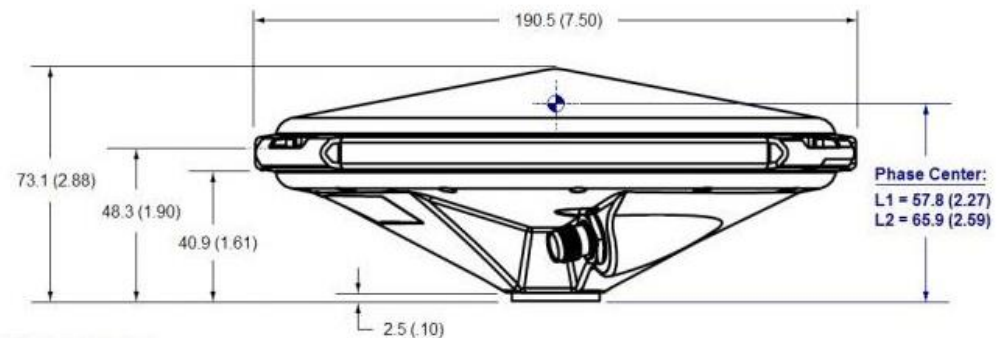


for a point;

- Conn receiver
- Conn receiver frequency;



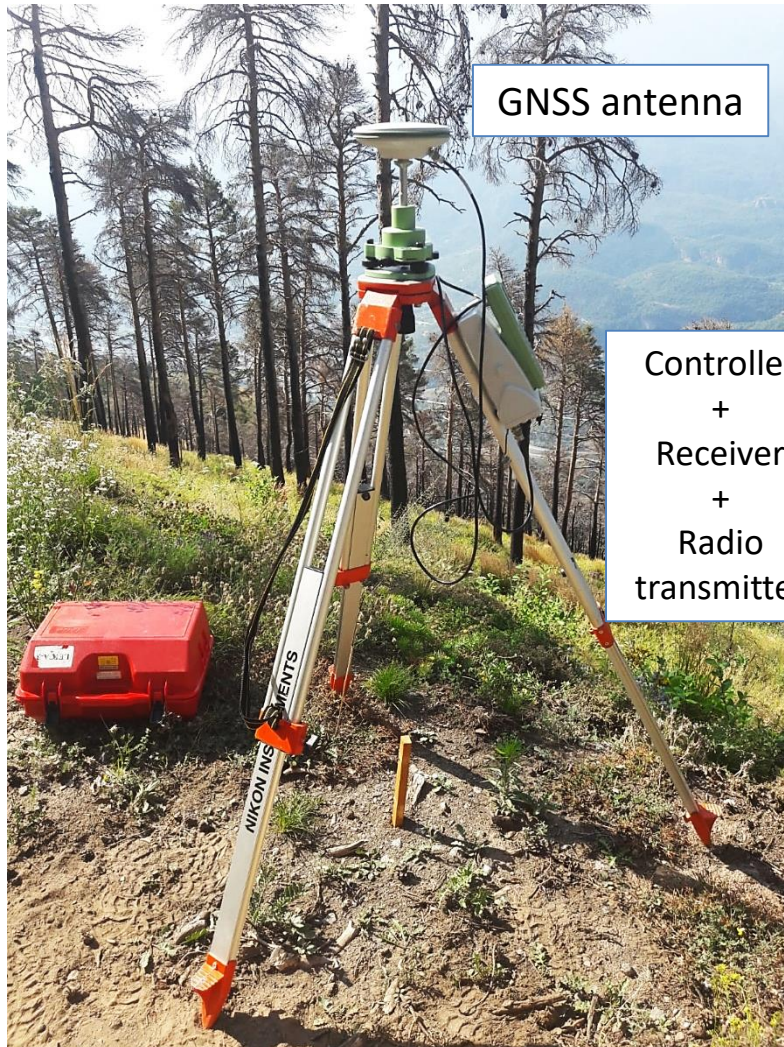
Phase Center – Survey GNSS Antenna (GESXGNSSANT31)





MASTER STATION

GNSS RTK positioning



GNSS antenna

Controller
+
Receiver
+
Radio
transmitter

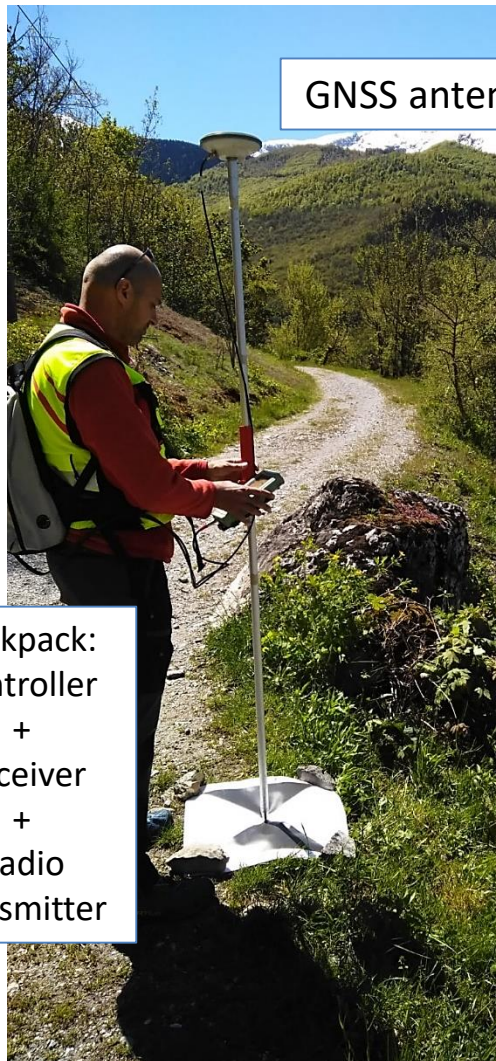
Operational steps

- Set up the tripod over a point;
- Measure the instrument height;
- Connect the GNSS antenna to the receiver;
- Connect the radio transmitter to the receiver and select the correct frequency;
- Start the receiver and choose the correct job settings;
- Set the known coordinate of the point;
- Start the data acquisition/transmission.



GNSS RTK positioning

ROVER STATION



GNSS antenna

Backpack:
Controller
+
Receiver
+
Radio
transmitter

Operational steps

- Set the **pole height**;
- Connect the **GNSS antenna** to the receiver;
- Connect the **radio transmitter** to the receiver and select the correct frequency;
- Start the receiver and choose the correct job settings;
- Set the known coordinate of the point;
- Place the instrument on the point and **“center the bubble”**;
- Start the data acquisition.

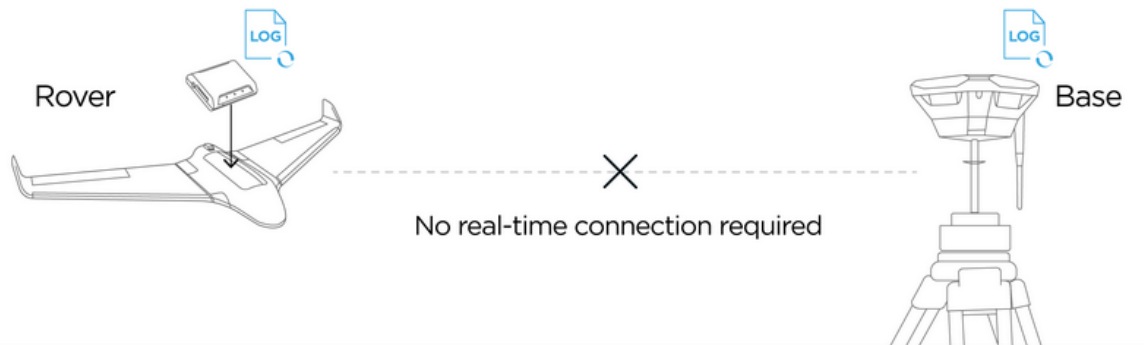


PPK positioning

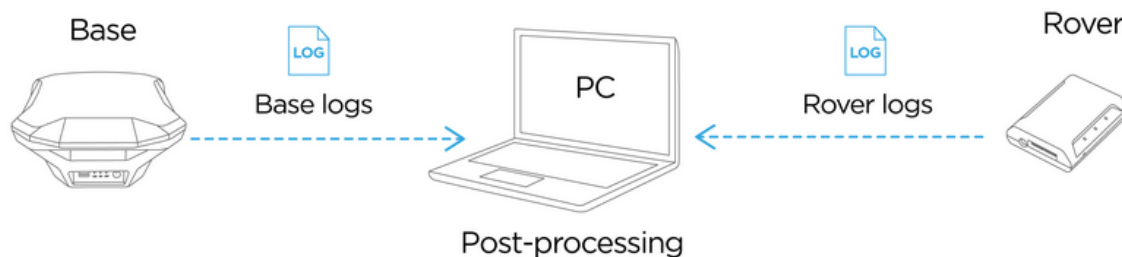
- Very high productivity

Is an alternative technique to RTK. Accurate positioning doesn't happen in real time. Both base on the ground and rover (usually on a UAV) record raw GNSS logs, which are then processed to receive an accurate positioning track.

① In the field

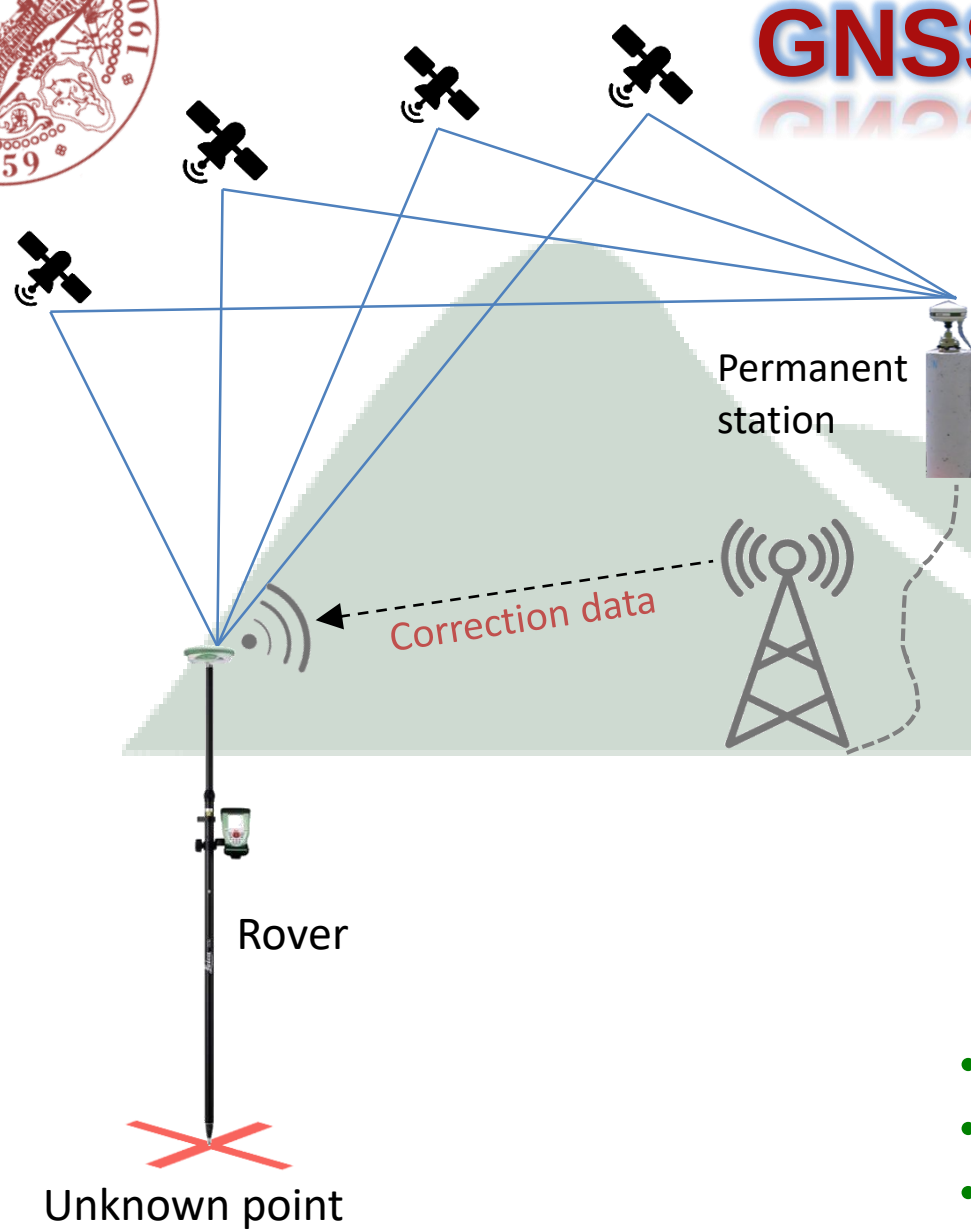


② In the office





GNSS NRTK positioning



Equipment: a dual-frequency geodetic receiver;

GSM / GPRS telephone modem (often integrated into the receiver or controller) that sends the position to the network in NMEA format and acquires the correction from the network in RTCM format.

- Real time solution
- Measurement time: 5-10 sec.
- High productivity



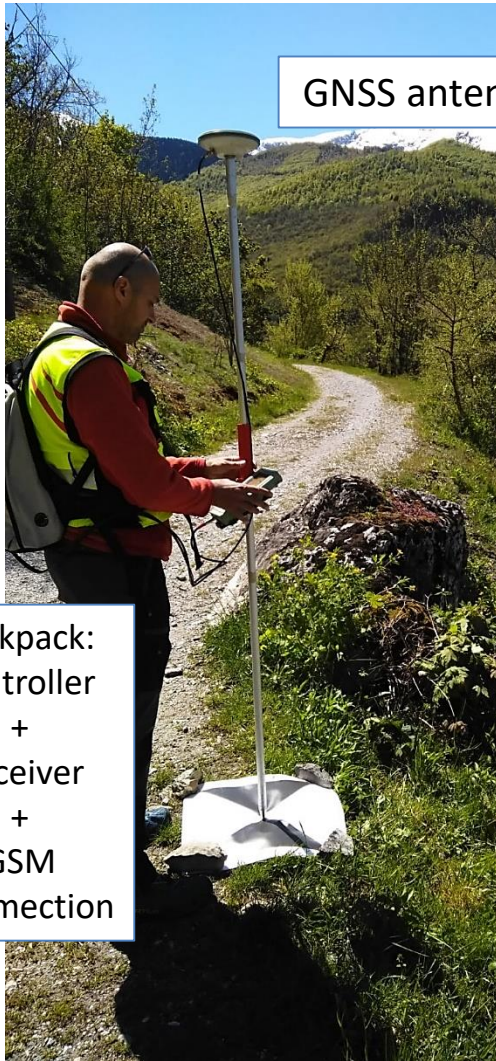
GNSS NRTK positioning

GNSS antenna

Operational steps

- Verify the coverage of the **network of permanent station**;
- Verify internet coverage;
- Set the **pole height**;
- Connect the **GNSS antenna** to the receiver;
- Turn on internet data service;
- Start the receiver and choose the correct job settings;
- Set the known coordinate of the point;
- Place the instrument on the point and **“center the bubble”**;
- Start the data acquisition.

Backpack:
Controller
+
Receiver
+
GSM
connection





SPIN GNSS Service

<https://www.spingnss.it> → **CORS Network** (Continuously operating reference Service)



SPIN3 GNSS

Servizio di Posizionamento Interregionale GNSS
Lombardia - Piemonte - Valle d'Aosta



Regione
Lombardia



REGIONE
PIEMONTE

Mapa dei Siti

SpiderWeb

Homepage

- ↓ Servizi della rete
- ↓ Stazioni della rete
- ↓ Mappa dei Siti
- ↓ Registrazione
- ↓ Recupera Password
- ↓ Centro Servizi
- ↓ Assistenza
- ↓ Domande e Risposte
- ↓ Download RINEX 30 s
- ↓ Guide e Utility
- ↓ Utilizzo dei dati di navigazione

Login

ID Utente*:

Password*:

☐ Ricordamelo

*Campo obbligatorio

Servizio di Posizionamento Interregionale GNSS
di Regione Piemonte, Regione Lombardia
e Regione Autonoma Valle d'Aosta

Si comunica agli utenti che il servizio SPIN 3 GNSS è nuovamente disponibile.



ATTENZIONE

per il collegamento RTK è necessario modificare l'indirizzo del server con il
nuovo IP: 158.102.7.10

Per qualsiasi informazione, contattare il servizio di assistenza nelle giornate lavorative.



Service responsible for
the acquisition, storage
and processing of code
and phase data deriving
from satellite
constellations, operating
24 hours a day for 365
days a year

Products:

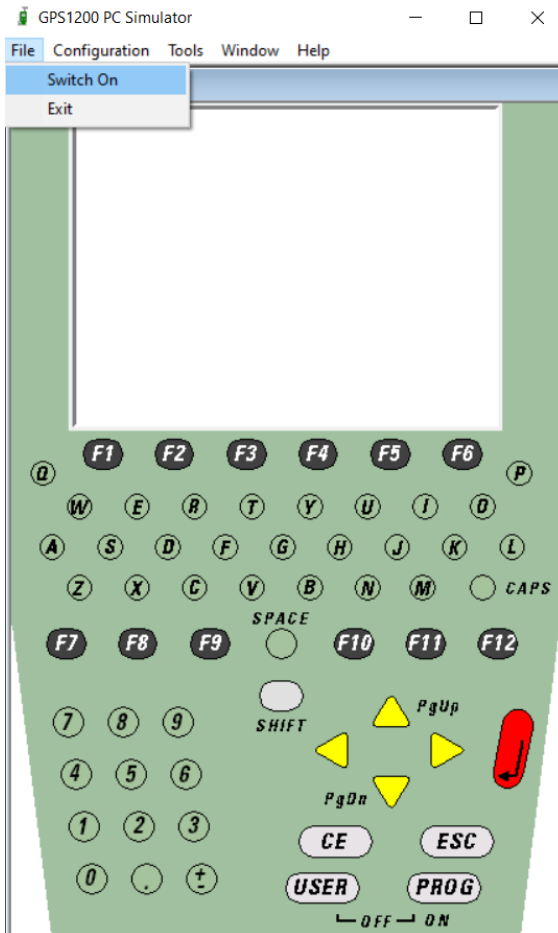
- RINEX (from 1" to 60")
for post-processing
- Virtual RINEX
- Real-time RTCM code
and phase



Ex. 1 – NRTK survey simulation

GPS1200 _RX1250 Simulation.zip

1. Turn on the receiver





Ex. 1 – NRTK survey simulation

GPS1200_RX1250 Simulation.zip

The screenshot displays four windows from the NRTK simulation software, illustrating the configuration process for a survey simulation. Red arrows indicate the sequence of steps:

- Interfaces Window (11:38):** Shows the configuration for the Real-Time interface. The 'Interface' is set to 'Real-Time', 'Port' is '1', and 'Device' is 'Siemens MC45'. The 'EDIT' button is highlighted.
- Connessione GSM Window (11:55):** Shows the GSM connection settings. 'Tipo GSM' is 'Manufact ModelId', 'Stazione' is 'TORINO', 'Numero' is '011-5647678', 'Protocollo' is 'Analog', 'AutoCONNEX' is 'No', and 'Net Data Rate' is '9600bps'.
- Real-Time Mode Window (11:42):** Shows the Real-Time Mode settings. 'R-Time Mode' is 'Rover', 'R-Time Data' is 'RTCM 18,19 v2', 'Port' is 'Port 1', 'Device' is 'Siemens MC45', 'Ref Sensor' is 'GX1230', and 'Ref Antenna' is 'Unknown'.
- Additional Rover Options Window (11:43):** Shows the 'General' tab of the 'Additional Rover Options' dialog. 'Accept Ref' is 'Any Received', 'Ref Stn ID' is '0', 'Ref Network' is 'Nearest', 'Send User ID' is 'i-MAX', 'User ID 1' is 'MAX', and 'User ID 2' is 'VRS'.



Ex. 1 – NRTK survey simulation

GPS1200 _RX1250 Simulation.zip

The sequence of screenshots shows the following steps:

- 11:57 CONFIGURE**: The 'Interfaces' window is shown. The 'Internet' interface is selected, showing 'Port 1' and '<Port 1>'.
- 12:01 CONFIGURE**: The 'Real-Time Mode' window is shown. 'R-Time Mode' is set to 'Rover', 'R-Time Data' is 'RTCM', 'Port' is 'Net 1', and 'Device' is 'Internet'.
- 12:02 CONFIGURE**: The 'Additional Rover Options' window is shown. 'Accept Ref' is set to 'Any Received'.
- 12:04 CONFIGURE**: The 'Set NET Port' window is shown. 'Name' is 'Net 1', 'User' is 'Client', 'IP Address' is '192.168.0.1', 'TCP/IP Port' is '0', and 'Auto CONEC' is 'No'.
- 12:04 CONFIGURE**: The 'Set NET Port' window is shown again, with the 'PAGE' button highlighted.

Red arrows indicate the flow from the 'Internet' interface selection to the 'Real-Time Mode' settings, then to the 'Additional Rover Options', and finally to the 'Set NET Port' configuration.



Ex. 1 – NRTK survey simulation

GPS1200 _RX1250 Simulation.zip

11:48
CONFIGURE

Devices

Name	Type
<Port 1>	<Port 1>
RS232	RS232
Smartgate	SMARTgate

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11:48
CONFIGURE

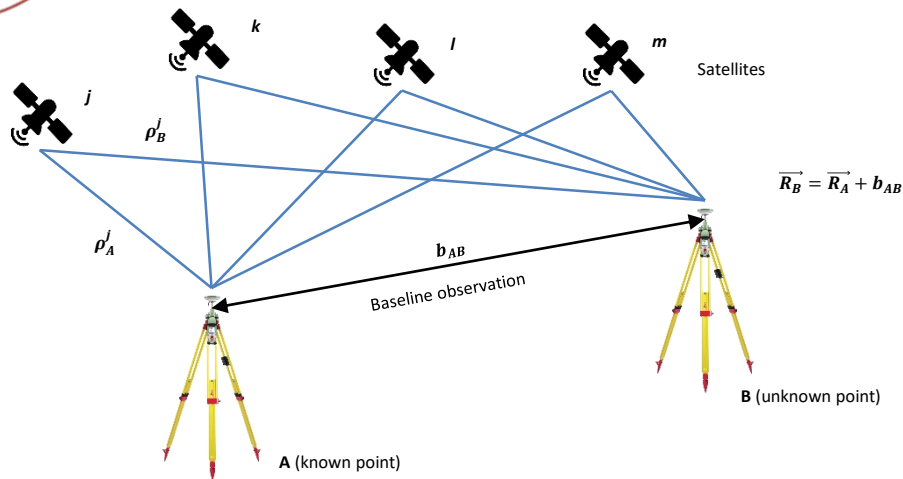
Devices

Name	Type
Modem	Modem
Motorola E1000	GSM
MultiTech CDMA	CDMA
Nokia 6230	GSM
Siemens M20	GSM
Siemens MC45	GSM
Siemens S25/S35i	GSM

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GNSS Static Network



Based on relative static or relative rapid-static positioning

Method used for the surveys of the **geodetic and topographical networks**, for **monitoring** of landslides, top of buildings, deformations, and any other phenomenon that involves the detection of movements.

- **High precision (mm)**
- **Post-processing of data** is necessary with modeling of systematic effects, solution of the ambiguity issue and rigorous definition of the reference system.
- Operating with N tools in this configuration we will obtain N -1 independent baselines of the network



PLANNING

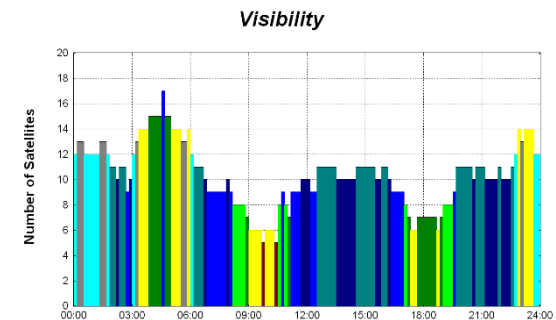
GNSS Static Network



Duration of measurement sessions.

Depend on:

- Final precision required;
- Length of the baselines;
- Satellite configuration (therefore presence of obstacles) during each session;
- Quality of the signal received at the various points.





PLANNING

To control and compensate the network, redundant measurements are necessary.

Network of **N points** -> **N-1 independent baselines** of the network, which do not form a closed polygon

$$N_{max} = \frac{N \cdot (N - 1)}{2}$$

$$R = \frac{N_0}{(N - 1)}$$

N_0 = Number of the measured baselines

R = redundancy factor

Having established the N_R number of receivers to be used and the redundancy required for the survey, it is necessary to plan the number, sequence and distribution of the observation sessions.

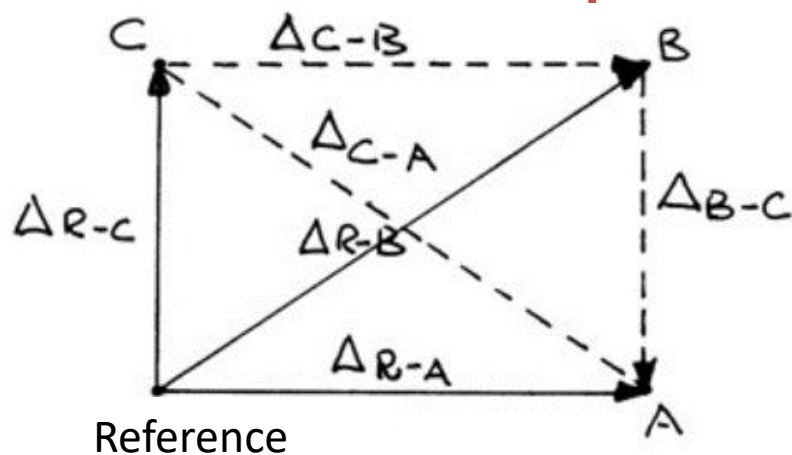
The total number of sessions to be carried out is therefore:

$$S = \frac{R \cdot (N - 1)}{(N_R - 1)}$$



GNSS Static Network

PLANNING - example



1 Reference station

Total of 4 point to occupy:

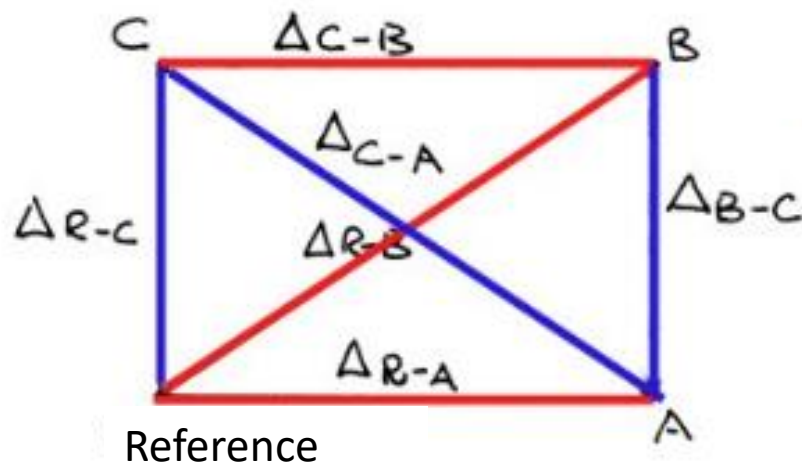
$$N_{\max} = 6 \text{ baselines}$$

Needed measures:

$$N-1 = 3 \text{ baselines}$$

Receivers	Baselines	Sessions
2	1	6
3	2	3
4	3	2

Solution



4 receivers,

2 measurement sessions,

3 independent baselines for each session